

ESIM Wiring Test

2018 Stelvio 2.0T | Does the ESIM switch signal reach the ECU intact?

Do first: wiTECH **flash check**, then **Test A** (vacuum on the ESIM while watching the ECU's switch reading). If Test A passes, the wiring is good. Run this sheet only if Test A never changes or flickers. No ESIM circuit code (e.g. P24D6) has set, so look for **intermittent / high-resistance** faults rather than a clean break.

Safety: back-probe only (never spread terminals); battery negative OFF before unplugging the ECU; fused jumper; fuel vapour at the canister, no sparks.

From the diagram (fill in first)

ESIM connector / signal pin, colour	Ground pin, colour / ground point	ECU connector + pin / reference V

Steps

- **B1 Connector:** ignition OFF, unplug the ESIM (driver-side rear wheel well). Look for corrosion, water, bent or pushed-back pins, loose terminal grip. Check the harness for chafe on the wheel-well edge.
- **B2 Signal reference:** connector unplugged, ignition RUN. Harness-side signal pin to chassis ground: expect the diagram's reference voltage. 0 V = open or ECU not driving it. Battery voltage where a low reference is expected = short to power.
- **B3 Ground:** harness-side ground pin to battery negative: near 0 V (ignition RUN) and a few ohms or less (ignition OFF).
- **B4 End to end:** battery negative disconnected, ECU connector unplugged. Each wire ESIM pin to ECU pin: a fraction of an ohm. Several ohms = high resistance; OL = break. Each wire to ground and to B+: open. Reconnect ECU, battery last.
- **B5 Jumper:** battery on, ignition RUN, ESIM unplugged, wiTECH ECM data display open. Fused jumper from signal pin to ground pin (= closed switch). wiTECH should show **closed**. Wiggle the harness along its route while watching.

Result

Finding	Meaning	Next
B5 shows closed, Test A failed	Wiring and ECU input good	ESIM switch/contacts: bench test or replace
B5 stays open	Wiring or ECU input	Jumper at the ECU end to split wire from ECU
B2/B3/B4 out of range	Open, short or high resistance	Repair that circuit (FCA terminal/splice)
Flicker on wiggle	Intermittent connection	Repair connector or chafe point

Record

Test	Result	Test	Result
B1 connector		B4 signal wire ohms / to gnd / to B+	
B2 signal reference V		B4 ground wire ohms	
B3 ground V / ohms		B5 wiTECH state / wiggle	

After a repair: clear codes, drive several days (fuel 15-85 %, cold starts), then cuore verify_repair for P0455/P0456/P0440 ('passed since clear' on all three). The check-engine light cannot confirm it.